**1** A ball of mass 130 g is launched from the ground at an initial velocity of 14 m / s vertically upwards. It decelerates until it is at rest momentarily at a height *h* above the ground.

**(a)** Define deceleration.

.....

..... [2]

**(b)** The acceleration of free fall is 9.8 m / s$^2$.

Show that the time taken for the ball to reach height *h* is 1.4 s. Ignore the effect of air resistance.

[1]

**(c)** Calculate *h*. Ignore the effect of air resistance.

*h* = ..... [3]

**(d)** The ball is dropped from the top of a tall building.

Describe and explain the motion of the ball as it falls. Consider the effect of air resistance in your answer.

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.....

.....

..... [3]

[Total: 9]

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